

Bloomberg Fixed Income LQA (Enhanced GSAC) — First Principles Explainer

A section-by-section rebuild of the methodology, with worked numeric examples.

GSAC = **G**overnment, **S**upranational, **A**gency, **C**orporate bonds. ~175,000 securities covered.

0. Why fixed income needs a *completely different* model from equities

The equity model (see companion document) works because equities trade on **continuous, transparent limit order books** with dense, high-frequency data — you can directly observe the order book, participation rates, and impact decay tick by tick.

Bonds are the opposite: - Most bonds **trade a handful of times a month or less** (“lack of trading activity for some securities and regions”). - There is **no central limit order book** — trading is dealer-intermediated, over-the-counter, quote-driven. - There is **no standardized clock-time measure of “how fast can I trade”** the way there is for equities — you can’t watch a continuously-refilling order book.

So the FI model **cannot** be built the equity way (observe impact decay tick-by-tick, fit a decay kernel). Instead, it takes a step back and asks a more primitive question: **at any moment, for any bond, how much volume is “available” to trade at a given acceptable cost, within a given time window — and how does that available volume behave statistically?** This is modeled as a **random variable**, not backed out from a deterministic impact curve. That’s the core philosophical difference from the equity model, and it’s why the FI paper’s math looks completely different (Gamma distributions and DNNs, rather than power-law decay kernels).

1. Core Concepts and Notation (Section 3.1)

Symbol	Meaning	Observable?
V	Trade Volume — the size of an actual/target execution	Yes (it’s what you choose)
AV	Available Volume — the largest size executable at a given cost, at a given moment	No — this is the hidden random variable the whole model exists to characterize

Symbol	Meaning	Observable?
C	Trade Cost — cost of executing, as % of price	Observed after the fact per trade
T	Trade Horizon — time to execute	Observed after the fact per trade (with adjustments)

The governing common-sense principle (stated directly in the paper): *liquidity increases as AV and T increase, and decreases as C decreases requirements tighten* — i.e. the more you’re willing to pay, and the more time you allow, the more volume becomes “available.” This single monotonicity is what the whole functional-form design (Sections 3.2, 4.1, 4.2) has to respect.

2. The Gamma Process — the foundational modeling choice (Section 3.2)

2.1 Why a Gamma distribution?

Available volume AV must be: - **Always non-negative** (you can’t have negative available volume). - **Right-skewed** — most of the time available volume is modest, but occasionally (fresh issuance, big dealer axe, momentum day) a lot of volume is available. - **Flexible in shape** — sometimes tightly clustered near a typical value, sometimes very dispersed, depending on the bond and market regime.

The **Gamma distribution**, $\text{Gamma}(k, \theta)$ — shape k , scale θ — satisfies exactly this: it’s supported on $(0, \infty)$, and by varying k you smoothly interpolate from a sharp exponential-like decay ($k = 1$) to something bell-shaped and roughly symmetric (large k), while θ just rescales the whole thing (like a “units” knob).

2.2 The defining equation (Eq. 3.1 — the single most important line in the paper)

$$AV_{C,T} \sim \text{Gamma}(k(C, T), \theta)$$

In words: for any chosen cost tolerance C and time window T , the volume you could actually get done is Gamma-distributed, where the **shape parameter k is itself a function of C and T** (grows as either grows, matching the “liquidity increases with C, T ” principle above), and θ is a per-bond scale constant (calibrated, roughly capturing the bond’s overall “size of a typical trade”).

This is analogous to (but structurally different from) the equity model’s approach: instead of writing a deterministic formula for expected cost and adding noise around it, the FI model writes a **distribution for volume** directly, and *derives* cost and horizon distributions from it (next section) — because in FI,

volume-at-a-price is the more primitive, more directly meaningful quantity (dealers quote “I can do \$2mm at this level,” not a smooth continuous impact curve).

2.3 Simple numeric intuition for the Gamma shape

A Gamma(k, θ) has mean $k\theta$ and variance $k\theta^2$ — so the **coefficient of variation is** $1/\sqrt{k}$: small $k \rightarrow$ very dispersed/uncertain available volume (typical of illiquid, thinly-traded bonds where k ends up small because cost/horizon inputs are modest); large $k \rightarrow$ more predictable available volume (liquid governments/agencies where even modest cost/horizon tolerances unlock a lot of relatively stable trading capacity).

k	θ	Mean AV	Std Dev AV	CV
0.5	2,000,000	1,000,000	1,414,214	141%
2	2,000,000	4,000,000	2,828,427	71%
8	2,000,000	16,000,000	5,656,854	35%

Same θ , but as k rises (more cost/time tolerance, or a bigger/older/more liquid bond), the *mean* available volume rises **and** it becomes proportionally *more predictable* — exactly the intuitive behavior you’d want.

3. From the Gamma Process to Horizon and Cost Distributions (Sections 3.3, 3.4)

This is pure probability mechanics once Eq. 3.1 is accepted — no new economic assumptions, just consequences of it.

3.1 Horizon distribution

Question: given I’m willing to pay up to cost $\bar{C}$ and want to move $\bar{V}$, what’s the distribution of the time T it will take?

Logic: $T \geq x$ (it takes at least x time) is equivalent to saying that the cumulative available volume, integrated up through time x at cost ceiling $\bar{C}$, hasn’t yet reached your target $\bar{V}$ (Eq. 3.3). Turning the crank through the Gamma CDF (regularized incomplete gamma function $\Gamma(\cdot, \cdot) / \gamma(\cdot, \cdot)$) gives (Eq. 3.6-3.7):

$$F_{T|\bar{V},\bar{C}}(x) = \frac{\Gamma(k(\bar{C}, x), \bar{V}/\theta)}{\Gamma(k(\bar{C}, x))}$$

This is exactly the **regularized upper incomplete Gamma function**, $Q(k(\bar{C}, x), \bar{V}/\theta)$ — a standard special function available in Excel as a formula (built from `GAMMA.DIST`).

3.2 Cost distribution

Symmetric logic for cost, holding volume and horizon fixed (Eq. 3.13-3.14):

$$F_{C|\bar{V},\bar{T}}(y) = Q\left(k(y, \bar{T}), \bar{V}/\theta\right)$$

This is the workhorse formula for the entire model — plug in any cost level y , your target volume $\bar{V}$, and your available horizon $\bar{T}$, and out comes the probability that you *could* execute at cost $\leq y$. Flip it around (find y such that this equals your desired confidence α) and you get the **cost at a given confidence level** — this is exactly `LQA_LIQUIDATION_COST` at a chosen `LQA_TGT_COST_CONF_LEVEL`.

3.3 Worked numeric example — Cost distribution from the Gamma model

Take a hypothetical corporate bond with calibrated $\theta = \$1,000,000$ and, for a target horizon $\bar{T} = 1$ day and a trial cost level $y = 0.50\%$, suppose the functional form (Section 4, next) gives $k(0.50\%, 1 \text{ day}) = 3.0$.

Target volume to trade: $\bar{V} = \$2,000,000$, so $\bar{V}/\theta = 2.0$.

$$F_C(0.50\%) = Q(3.0, 2.0) = 1 - P(3.0, 2.0)$$

Using the regularized lower incomplete gamma function $P(3, 2) \approx 0.323$ (standard tables / Excel `GAMMA.DIST(2, 3, 1, TRUE)`), so $Q(3, 2) \approx 0.677$.

Interpretation: there's about a 67.7% chance you can execute \$2mm within 1 day at a cost of 0.50% or less. If you need higher confidence (say 90%), you'd need to either accept a higher cost ceiling y (which raises $k(y, T)$ further and pushes Q up) or allow more time T (same effect via the T dependence baked into $k(C, T)$).

(The companion Excel workbook's "FixedIncome" tab makes this fully formula-driven — flex $\bar{V}$, $\bar{T}$, y and watch k , Q , and the implied confidence recompute live.)

4. The Functional Forms — $k(\cdot)$ and $s(\cdot)$ (Section 4.1-4.2)

Section 3 tells you *what* $k(C, T)$ does (drives the Gamma shape); Section 4 tells you its actual *formula*:

$$k(C, T) = T \cdot s\left(k_1, k_2, \frac{C}{\text{Volatility}^{k_3}}\right) \cdot \frac{\text{Amt}}{\theta} \quad (\text{Eq. 4.1})$$

where

$$s(k_1, k_2, x) = \frac{k_1 x}{k_1 x + \exp(k_2 - k_1 x)} \quad (\text{Eq. 4.2, a sigmoid})$$

Take this apart term by term:

1. **T (linear in horizon):** the shape parameter scales up linearly with how much time you allow — directly matching “available volume grows over time” (more time = proportionally more shape = fatter/higher-mean Gamma).
2. $s(k_1, k_2, C/\text{Volatility}^{k_3})$ — **a bounded sigmoid in volatility-normalized cost:** the input to the sigmoid is cost C scaled down by $(\text{price volatility})^{k_3}$ — so for a *given raw cost tolerance* C , a more volatile bond effectively has *less* real cost tolerance in this normalized sense (its price moves more on its own, so a fixed % cost buffer buys you less “real” liquidity access). The sigmoid $s(\cdot)$ is bounded between 0 and 1 — this directly encodes **diminishing returns / concavity**: past a certain (volatility-adjusted) cost tolerance, offering to pay even more doesn’t unlock proportionally more volume. This is exactly the Abstract’s claim that “*market impact is concave on large order sizes.*”
3. Amt/θ **(linear in amount outstanding, scaled):** bigger bond issues (more amount outstanding) have proportionally more available volume — a \$2 billion issue has more capacity than a \$200 million issue, all else equal. This also matches the SHAP feature-importance finding (Section 5.1, Fig. 5) that **Amount Outstanding is the #1 or #2 most important feature in every one of the four sub-models.**

4.1 Worked numeric example — computing $k(C, T)$

Hypothetical corporate bond: annualized price volatility = 8% ($k_3 = 1$, so $\text{Volatility}^{k_3} = 0.08$), Amount Outstanding = \$500,000,000, $\theta = \$1,000,000$ (calibrated), sigmoid parameters $k_1 = 40$, $k_2 = 1.5$ (illustrative).

For a cost tolerance $C = 0.50\% = 0.005$ and horizon $T = 1$ day:

$$\begin{aligned}
 x &= C/\text{Volatility}^{k_3} = 0.005/0.08 = 0.0625 \\
 s(40, 1.5, 0.0625) &= \frac{40 \times 0.0625}{40 \times 0.0625 + \exp(1.5 - 40 \times 0.0625)} = \frac{2.5}{2.5 + \exp(1.5 - 2.5)} = \frac{2.5}{2.5 + 0.3679} = 0.8718 \\
 k(0.5\%, 1) &= 1 \times 0.8718 \times \frac{500,000,000}{1,000,000} = 0.8718 \times 500 = 435.9
 \end{aligned}$$

(This k is illustrative only — the paper’s actual k_1, k_2, k_3, θ are DNN-estimated per bond, not disclosed numerically; the Excel workbook lets you experiment with your own plausible parameter choices and see how sensitive k — and downstream cost/horizon distributions — are to each one.)

5. Market Risk Adjustment (Sections 4.3-4.5)

The problem: the price can move *between* when you decide to trade and when the trade actually executes, for reasons that have nothing to do with your own liquidity-consumption. This “market risk” component can make realized cost *negative* (favorable price move) or *unusually large* (unfavorable move) relative to the “true” liquidity cost $\tilde{C}$.

The constraint: true liquidation cost $\tilde{C} \geq 0$ always (a dealer never pays *you* a premium purely for the privilege of providing you liquidity) — so all the negativity/dispersion in *observed* cost is attributed to the error term ϵ :

$$C_{i,t} = \tilde{C}_{i,t} + \epsilon_{i,t}, \quad \epsilon_{i,t} < C_{i,t}$$

ϵ is modeled as **truncated normal** (Eq. 4.7) — normal, but truncated so that $\tilde{C} = C - \epsilon \geq 0$ always holds:

$$f_{\epsilon|C}(\epsilon, C) = \frac{\varphi(\epsilon/\sigma)}{\sigma \cdot \Phi(C/\sigma)}$$

— a standard normal density φ , rescaled by σ (market-risk std dev, e.g. from the bond’s price volatility over the horizon), and renormalized (divided by $\Phi(C/\sigma)$) so the truncated density still integrates to 1 over the allowed range.

The likelihood function used to fit the whole model (Eq. 4.5-4.6) integrates over this error term — i.e. **the model doesn’t get confused by market-risk noise in the training data; it explicitly separates “cost due to illiquidity” from “cost due to the market just moving.”** This is directly analogous to the equity model’s separation of the deterministic impact term $I(V, \tau)$ from the noise term $\sigma_E \sqrt{\tau/3} \cdot \epsilon$ — **both models insist on decomposing observed cost into a liquidity-driven piece and a market-risk piece**, just via different mechanics (log-normal multiplicative noise in equities; truncated-normal additive noise in FI).

Practical implication: the `LQA_MARKET_RISK_FLAG` (PY371) lets you toggle this on/off — off gives you the “pure liquidity cost” with market risk noise stripped out, useful when you want to isolate microstructure liquidity cost from directional price-risk exposure (e.g., for a liquidity stress test where you want to shock *liquidity*, not *market direction*, independently).

6. Confidence Levels (Section 4.4)

$$F(C_\alpha | \tilde{V}, \tilde{T}) = \alpha$$

C_α is simply the **quantile function** — invert the cost CDF from Section 3.2 at your chosen confidence α . Set $\alpha = 50\%$ for the median cost; $\alpha = 90\%$ for “the cost I won’t exceed 90% of the time” — a natural way to build in a safety margin for a stress-testing or worst-case-planning context, exactly analogous to the equity model’s `LQA_TGT_CONFIDENCE_LEVEL`.

7. Model Training — DNNs + SHAP (Section 5)

Rather than fitting (k_1, k_2, k_3, θ) once per market (as equities does with 7 regions), the FI model estimates a **unique set of these four parameters for every bond, at every point in time** — “amortized inference” via a neural network that maps each bond’s static + dynamic features (Tables 3-4: age, maturity, coupon, log(amount outstanding), recent trading activity ratios, sector dummies, callable/144A/green-bond flags, etc.) directly to (k_1, k_2, k_3, θ) .

- **Two hidden layers, ReLU activation, Adam optimizer** — standard, small feedforward network (not deep in the “many-layer” sense — this is a calibration surface fit, not an image-recognition-scale model).
- **SHAP analysis** picks the features that matter, separately per bond universe (USD IG, USD HY, Liquid Government-Related, Other Bonds) — and **Amount Outstanding + recent trading activity dominate in every universe**, confirming the economic intuition baked into Eq. 4.1.
- **Trades weighted 90%, quotes weighted 10%** in training — quotes fill in gaps for illiquid bonds (where trades are rare) but trades remain the ground truth where available.

Why this matters for you as a user: the model is not “one curve per bond” fit from that bond’s own sparse trade history (which would be hopeless for bonds that trade twice a year) — it’s a **cross-sectional model** that learns “bonds with these characteristics behave like this,” so even a newly-issued or rarely-traded bond gets a sensible liquidity estimate by virtue of resembling other bonds with richer trading histories.

8. Convertible Bonds — a special case (Section 6)

If a convertible bond is deep enough in-the-money (stock price > conversion price, delta > 80, within the conversion window), its liquidity is dominated by the *equity* it converts into, not its own thin bond-market trading — so the model **switches it to the equity LQA methodology** entirely. Otherwise it stays on the standard GSAC/Gamma-process model. This is a clean example of models needing to know their own boundaries: a bond model applied to a de-facto-equity instrument would badly *understate* its true liquidity.

9. Derived Outputs (Section 7)

9.1 Roundtrip cost (LQC)

$$LQC = \text{Liquidation cost to BUY} + \text{Liquidation cost to SELL}$$

for a **fixed reference size** — \$10 million for the most liquid govt/agency bonds, \$1 million for everything else — normalized by bid price, expressed in price units (0.0001 = 1bp = 0.01%). This gives a single comparable “round-trip friction” number per bond, in the spirit of Feldhütter (2012)’s search-friction literature.

9.2 Liquidity Score

A pure **1-100 percentile rank** of LQC *within the bond’s asset class* — 100 = cheapest to round-trip (most liquid) in that class, 1 = most expensive (least liquid). Purely relative — a score of 60 in Corporates and a score of 60 in Sovereigns do **not** mean the same absolute liquidity; they mean the same *relative standing within their own peer group* (LQA_SECTOR_SCORE , PY338, does the same within an even narrower “Liquidity Sector” peer group).

10. Model Performance / Backtesting (Section 8)

Three separate validation exercises, mirroring the three outputs: 1. **Expected Daily Volume** — predicted vs. observed mean, bucketed, checked against the 45° line (near-perfect alignment across all four bond universes). 2. **Liquidation Cost** — GSAC model residuals beat a naive bid-ask-spread-only benchmark; **Table 5** backtest stats (R^2 0.79-0.98, Spearman 0.86-1.00, Anderson-Darling p 0.32-0.72 — i.e., residuals are statistically consistent with being normally distributed, supporting good model fit) across all four sub-universes (USD IG, USD HY, Liquid Government-Related, Other Bonds). 3. **Liquidation Horizon** — a well-behaved, roughly linear relationship between predicted and observed trade windows across volume buckets (\$1mm/\$2mm/\$5mm), with reduced- χ^2 diagnostics (**Table 6**) all reasonably close to 1 (best fit), with the \$5mm bucket sitting right at the edge of the critical threshold (1.05 vs ~1.038) — the model’s weakest (but still broadly acceptable) spot is very large trade sizes, which is intuitive (least data density at the tails).

11. Appendix C's Worked Examples — a *pedagogical simplification*, not the production model

Important distinction to keep straight: Appendix C's four examples (reproduced in full below) use a small, discrete, empirical “sample dataset” of just four rows of (LV, LC, LH, count-of-observations) and compute conditional expectations by simple weighted averaging. **This is not literally how the production Gamma-process model computes its outputs** — it's a simplified, intuition-building illustration of the *general principle* (“condition on two of {cost, horizon, volume} to get the expected value / distribution of the third using the empirical joint likelihood”), using toy discrete counts instead of the continuous Gamma machinery from Sections 3-4. Think of it as the FI paper's way of saying “*here's the conceptual shape of a conditional-expectation calculation, in a form simple enough to verify by hand — the real model just replaces this discrete table with a smooth, DNN-calibrated Gamma joint density.*”

Sample dataset (Table 9):

LV (Million)	LC (%)	LH (Days)	Count
1	0.50	1	1
1	0.40	10	250
2	0.45	10	750
1	0.50	10	250

Example 1 — expected LV given LC=0.50%, LH=1 day: only one row matches (LV=1, LC=0.50, LH=1, count=1) → $E[LV]=1$ million (trivial, single observation).

Example 2 — expected LH given LC=0.50%, LV=1 million: two rows match LC=0.50 (regardless of LH): the LH=1 row (count 1) and the LH=10 row (count 250) — total 251 observations.

$$P(LH = 1) = 1/251, \quad P(LH = 10) = 250/251$$

$$E[LH \mid LC = 0.5, V = 1] = \frac{1}{251}(1) + \frac{250}{251}(10) = 9.9641 \text{ days}$$

Example 3 — expected LC given LV=1 million, LH=1 day: only the first row matches → $E[LC]=0.50\%$ (trivial, single observation).

Example 4 — expected LC given LV=1 million, LH=10 days: two rows match (LV=1, LH=10): LC=0.40 (count 250) and LC=0.50 (count 250) — 500 observations total.

$$P(LC = 0.40) = 250/500, \quad P(LC = 0.50) = 250/500$$

$$E[LC \mid LV = 1, LH = 10] = \frac{250}{500}(0.40) + \frac{250}{500}(0.50) = 0.45\%$$

(The Excel workbook’s “FixedIncome” tab reproduces this exact table and both formulas — SUMPRODUCT -based conditional expectations — as a sanity-checkable building block, right alongside the continuous Gamma-model calculations from Sections 2-4 above, so you can see both the toy discrete version and the real continuous version side by side.)

12. Model Inputs & Outputs (Appendix A/B)

Key outputs (Appendix A): LQA_EXPECTED_DAILY_VOLUME (PY229), LQA_LIQUIDATION_VOLUME (PY248) ± uncertainty (PY249), LQA_LIQUIDATION_COST (PY251) ± uncertainty (PY252), LQA_LIQUIDATION_HORIZON (PY254) ± uncertainty (PY255), LQA_LIQUIDATION_CONFIDENCE (PY246), LQA_LIQUIDITY_SCORE (PY275), LQA_SECTOR_SCORE (PY338), LQC (PY406, roundtrip cost).

Key overrides (Appendix B): LQA_TGT_LIQUIDATION_HORIZON (PY253), LQA_BID_ASK_SPREAD (PY273, default cost basis), LQA_TGT_COST_CONF_LEVEL (PY328), LQA_MARKET_RISK_FLAG (PY371, Y/N), factor-scaling overrides for volume/volatility/spread (PY400-402), and trade direction (PY472, Buy/Sell).

13. How this is used in practice — liquidity risk modeling for bond portfolios

1. **Cash-raising / redemption stress testing:** for a fund holding GSAC bonds (govies, agencies, IG/HY corporates), aggregate LQA_LIQUIDATION_HORIZON at a stressed confidence level (e.g. 90th percentile) across the whole book to answer “how many days, at what cost, to raise \$X of cash under adverse conditions” — directly analogous to the equity use case, and combinable with it for multi-asset-class funds.
2. **Liquidity bucketing for regulatory reporting** (e.g. UK/EU liquidity risk frameworks, US 22e-4-style classifications): LQA_LIQUIDITY_SCORE / LQA_SECTOR_SCORE give a systematic, model-based way to bucket positions into liquidity tiers instead of relying on static rules-of-thumb (e.g. “all IG corporates = liquid”).
3. **New-issue / off-the-run bond liquidity assessment:** because the model is cross-sectional (DNN-driven, not per-bond-history-driven), it can sensibly estimate liquidity for bonds with almost no trading history — directly useful for onboarding new issuance into a portfolio or assessing private-placement-adjacent or thinly-traded corporate bonds, which is exactly the kind of gap analysis relevant to private credit / direct lending risk teams who need *some* defensible liquidity proxy even for instruments that don’t trade like public benchmarks.

4. **Roundtrip cost (LQC) as a standardized transaction-cost budgeting input:** for portfolio construction / rebalancing cost-benefit analysis, LQC gives a comparable “cost to trade” number across very different bond types (a govt vs. a HY corporate) on a single normalized scale.
5. **Separating market risk from liquidity risk (the market-risk flag):** when stress-testing, you can isolate *pure liquidity deterioration* (spreads widening, available volume shrinking) from *directional price risk* (rates/credit spreads moving) — important for attribution, since these are genuinely different risk factors that a risk committee will want disentangled.

The one-line mental model to carry forward: *fixed income liquidity is modeled as a hidden, Gamma-distributed “available volume” that grows with how much cost and time you’re willing to tolerate and with how big the bond issue is; a neural network learns, per bond, exactly how fast that growth happens from the bond’s own characteristics — and every cost/horizon output you see on the terminal is just a slice of that one underlying Gamma distribution.*

Where the gap-filling agents flagged genuine uncertainty (for completeness)

- Two figure captions (Figure 9 “Probability density of cost residuals,” and Figure 10’s caption in Section 8.3) were listed in the paper’s List of Figures but their exact wording wasn’t captured on video — their content is inferred by close analogy to neighboring figures (Fig. 8 and Fig. 11 respectively) but is not verbatim.
- The Table of Contents’ exact page-number layout (pp.3-4) suffered from motion-blur double-exposure in the source video and is reconstructed from context rather than directly read.
- Every formula, every table’s numeric body, and the full reference list were otherwise directly, visually confirmed across two rounds of frame sampling — including the two most important equations in the paper (Eq. 3.1, the Gamma-process definition, and Eq. 4.1, the functional form of k), which required a second, denser sampling pass to recover.